Chapter 25

Basel I, Basel II, and Solvency II

An agreement in 1988, known as the Basel Accord, marked the start of international standards for bank regulation. Since 1988, bank regulation has been an evolutionary process. New regulations have modified previous regulations, but often approaches used in previous regulations have been preserved. In order to understand the current regulatory environment, it is therefore necessary to understand historical developments. This chapter explains the evolution of the regulatory environment prior to the 2007 Global Financial Crisis. Chapters 26 and 27 will cover developments since the crisis.

This chapter starts by explaining the 1988 Basel Accord (now known as Basel I), netting provisions, and the 1996 Amendment. It then moves on to discuss Basel II, which is a major overhaul of the regulations and was implemented by many banks throughout the world in about 2007. Finally, it reviews Solvency II, a regulatory framework for insurance companies, which is broadly similar to Basel II and was implemented by the European Union in 2016.

25.1 The Reasons for Regulating Banks

The main purpose of bank regulation is to ensure that a bank keeps enough capital for the risks it takes. It is not possible to eliminate altogether the possibility of a bank failing, but governments want to make the probability of default for any given bank very small. By

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Systemic Risk

Systemic risk is the risk that a default by one financial institution will create a "ripple effect" that leads to defaults by other financial institutions and threatens the stability of the financial system. There are huge numbers of over-the-counter transactions between banks. If Bank A fails, Bank B may take a huge loss on the transactions it has with Bank A. This in turn could lead to Bank B failing. Bank C that has many outstanding transactions with both Bank A and Bank B might then take a large loss and experience severe financial difficulties; and so on.

The financial system has survived defaults such as Drexel in 1990, Barings in 1995, and Lehman Brothers in 2008 very well, but regulators continue to be concerned. During the market turmoil of 2007 and 2008, many large financial institutions were bailed out, rather than being allowed to fail, because governments were concerned about systemic risk.

doing this, they hope to create a stable economic environment where private individuals and businesses have confidence in the banking system.

It is tempting to argue: "Bank regulation is unnecessary. Even if there were no regulations, banks would manage their risks prudently and would strive to keep a level of capital that is commensurate with the risks they are taking." Unfortunately, history does not support this view. There is little doubt that regulation has played an important role in increasing bank capital and making banks more aware of the risks they are taking.

As discussed in Section 2.3, governments provide deposit insurance programs to protect depositors. Without deposit insurance, banks that took excessive risks relative to their capital base would find it difficult to attract deposits. However, the impact of deposit insurance is to create an environment where depositors are less discriminating. A bank can take large risks without losing its deposit base.[1] The last thing a government wants is to create a deposit insurance program that results in banks taking more risks. It is therefore essential that deposit insurance be accompanied by regulation concerned with capital requirements.

A major concern of governments is what is known as *systemic risk*. This is the risk that a failure by a large bank will lead to failures by other large banks and a collapse of the financial system. The way this can happen is described in Business Snapshot 25.1. When a bank or other large financial institution does get into financial difficulties, governments

[1] As explained in Chapter 3, this is an example of what insurance companies term moral hazard. The existence of an insurance contract changes the behavior of the insured party.

have a difficult decision to make. If they allow the financial institution to fail, they are putting the financial system at risk. If they bail out the financial institution, they are sending the wrong signals to the market. There is a danger that large financial institutions will be less vigilant in controlling risks if they know that they are "too big to fail" and the government will always bail them out.

During the market turmoil of 2008, the decision was taken to bail out some financial institutions in the United States and Europe. However, Lehman Brothers was allowed to fail in September 2008. Possibly, the United States government wanted to make it clear to the market that bailouts for large financial institutions were not automatic. However, the decision to let Lehman Brothers fail has been criticized because arguably it made the Global Financial Crisis worse.

25.2 Bank Regulation Pre-1988

Prior to 1988, bank regulators within a country tended to regulate bank capital by setting minimum levels for the ratio of capital to total assets. However, definitions of capital and the ratios considered acceptable varied from country to country. Some countries enforced their regulations more diligently than other countries. Increasingly, banks were competing globally and a bank operating in a country where capital regulations were lax was considered to have a competitive edge over one operating in a country with tighter, more strictly enforced, capital regulations. In addition, the huge exposures created by loans from the major international banks to less developed countries such as Mexico, Brazil, and Argentina, as well as the accounting games sometimes used for those exposures (see Business Snapshot 2.3) were starting to raise questions about the adequacy of capital levels.

Another problem was that the types of transactions entered into by banks were becoming more complicated. The over-the-counter derivatives market for products such as interest rate swaps, currency swaps, and foreign exchange options was growing fast. These contracts increase the credit risks being taken by a bank. Consider, for example, an interest rate swap. If the counterparty in the interest rate swap transaction defaults when the swap has a positive value to the bank and a negative value to the counterparty, the bank is liable to lose money. The potential future exposure from derivatives was not reflected in the bank's reported assets. As a result, it had no effect on the level of assets reported by a bank and therefore no effect on the amount of capital the bank was required to keep. It became apparent to regulators that the value of total assets was no longer a good indicator of the total risks being taken. A more sophisticated approach than that of setting minimum levels for the ratio of capital to total balance-sheet assets was needed.

The Basel Committee was formed in 1974. The committee consisted of representatives from Belgium, Canada, France, Germany, Italy, Japan, Luxembourg, the Netherlands, Sweden, Switzerland, the United Kingdom, and the United States. It met regularly in Basel, Switzerland, under the patronage of the Bank for International

Settlements. The first major result of these meetings was a document titled "International Convergence of Capital Measurement and Capital Standards." This was referred to as "The 1988 BIS Accord" or just "The Accord." Later it became known as Basel I.

25.3 The 1988 BIS Accord

The 1988 BIS Accord was the first attempt to set international risk-based standards for capital adequacy. It has been subject to much criticism as being too simple and somewhat arbitrary. In fact, the Accord was a huge achievement. It was signed by all 12 members of the Basel Committee and paved the way for significant increases in the resources banks devote to measuring, understanding, and managing risks. The key innovation in the 1988 Accord was the Cooke ratio.

25.3.1 The Cooke Ratio

The Cooke ratio[2] considers credit risk exposures that are both on-balance-sheet and off-balance-sheet. It is based on what is known as the bank's total *risk-weighted assets* (also sometimes referred to as the *risk-weighted amount*). This is a measure of the bank's total credit exposure.

Credit risk exposures can be divided into three categories:

1. Those arising from on-balance sheet assets (excluding derivatives)
2. Those arising for off-balance sheet items (excluding derivatives)
3. Those arising from over-the-counter derivatives

Consider the first category. Each on-balance-sheet asset is assigned a risk weight reflecting its credit risk. A sample of the risk weights specified in the Accord is shown in Table 25.1. Cash and securities issued by governments of OECD countries (members of the Organisation for Economic Co-operation and Development) are considered to have virtually zero risk and have a risk weight of zero. Loans to corporations have a risk weight of 100%. Loans to banks and government agencies in OECD countries have a risk weight of 20%. Uninsured residential mortgages have a risk weight of 50%. The total of the risk-weighted assets for N on-balance-sheet items equals

$$\sum_{i=1}^{N} w_i L_i$$

where L_i is the principal amount of the ith item and w_i is its risk weight.

[2]The ratio is named after Peter Cooke, from the Bank of England.

Table 25.1 Risk Weights for On-Balance-Sheet Items

Risk Weight (%)	Asset Category
0	Cash, gold bullion, claims on OECD governments such as Treasury bonds or insured residential mortgages
20	Claims on OECD banks and OECD public-sector entities such as securities issued by U.S. government agencies or claims on municipalities
50	Uninsured residential mortgage loans
100	All other claims such as corporate bonds and less developed country debt, claims on non-OECD banks

Example 25.1

The assets of a bank consist of \$100 million of corporate loans, \$10 million of OECD government bonds, and \$50 million of residential mortgages. The total of the risk-weighted assets is

$$1.0 \times 100 + 0.0 \times 10 + 0.5 \times 50 = 125$$

or \$125 million.

Consider next the second category. This includes bankers' acceptances, guarantees, and loan commitments. A credit equivalent amount is calculated by applying a conversion factor to the principal amount of the instrument. Instruments that from a credit perspective are considered to be similar to loans, such as bankers' acceptances, have a conversion factor of 100%. Others, such as note issuance facilities (where a bank agrees that a company can issue short-term paper on pre-agreed terms in the future), have lower conversion factors.

Finally, consider the third category. For an over-the-counter derivative such as an interest rate swap or a forward contract, the credit equivalent amount is calculated as

$$\max(V, 0) + aL \tag{25.1}$$

where V is the current value of the derivative to the bank, a is an *add-on factor*, and L is the principal amount. The first term in equation (25.1) is the current exposure. If the counterparty defaults today and V is positive, the contract is an asset to the bank and the bank is liable to lose V. If the counterparty defaults today and V is negative, the contract is an asset to the counterparty and there will be neither a gain nor a loss to the bank. The bank's exposure is therefore $\max(V, 0)$. (More details on the bilaterally cleared over-the-counter derivatives market are in Chapters 6 and 18. At this stage, netting was not recognized in the calculation of bank required capital.) The add-on amount, aL, is an allowance for the possibility of the exposure increasing in the future. Examples of the add-on factor, a, are shown in Table 25.2. Equation (25.1) is known as the *current exposure method* (CEM). The add-on factors were revised and extended in the years following 1988.

Table 25.2 Add-On Factors as a Percent of Principal for Derivatives

Remaining Maturity (yr)	Interest Rate	Exchange Rate and Gold	Equity	Precious Metals Except Gold	Other Commodities
< 1	0.0	1.0	6.0	7.0	10.0
1 to 5	0.5	5.0	8.0	7.0	12.0
> 5	1.5	7.5	10.0	8.0	15.0

Example 25.2

A bank has entered into a $100 million interest rate swap with a remaining life of four years. The current value of the swap is $2.0 million. In this case, the add-on amount is 0.5% of the principal so that the credit equivalent amount is $2.0 million plus $0.5 million or $2.5 million.

The credit equivalent amount arising from either the second or third category of exposures is multiplied by the risk weight for the counterparty in order to calculate the risk-weighted assets. The risk weights are similar to those in Table 25.1 except that the risk weight for a corporation is 0.5 rather than 1.0.

Example 25.3

Consider again the bank in Example 25.2. If the interest rate swap is with a corporation, the risk-weighted assets are 2.5×0.5 or $1.25 million. If it is with an OECD bank, the risk-weighted assets are 2.5×0.2 or $0.5 million.

Putting all this together, the total risk-weighted assets for a bank with N on-balance-sheet items and M off-balance-sheet items is

$$\sum_{i=1}^{N} w_i L_i + \sum_{j=1}^{M} w_j^* C_j \qquad (25.2)$$

Here, L_i is the principal of the ith on-balance-sheet asset and w_i is the risk weight for the asset; C_j is the credit equivalent amount for the jth derivative or other off-balance-sheet item and w_j^* is the risk weight of the counterparty for this jth item.

25.3.2 Capital Requirement

The Accord required banks to keep capital equal to at least 8% of the risk-weighted assets. The capital had two components:

1. *Tier 1 Capital.* This consists of items such as equity and noncumulative perpetual preferred stock.[3] (Goodwill is subtracted from equity.[4])
2. *Tier 2 Capital.* This is sometimes referred to as Supplementary Capital. It includes instruments such as cumulative perpetual preferred stock,[5] certain types of 99-year debenture issues, and subordinated debt (i.e., debt subordinated to depositors) with an original life of more than five years.

Equity capital is the most important type of capital because it absorbs losses. If equity capital is greater than losses, a bank can continue as a going concern. If equity capital is less than losses, the bank is insolvent. In the latter case, Tier 2 capital becomes relevant. Because it is subordinate to depositors, it provides a cushion for depositors. If a bank is wound up after its Tier I capital has been used up, losses should be borne first by the Tier 2 capital and, only if that is insufficient, by depositors. (See Section 2.2.)

The Accord required at least 50% of the required capital (that is, 4% of the risk-weighted assets) to be in Tier 1. Furthermore, the Accord required 2% of risk-weighted assets to be common equity. (Note that the Basel Committee has updated its definition of instruments that are eligible for Tier 1 capital and its definition of common equity in Basel III.)

The bank supervisors in some countries require banks to hold more capital than the minimum specified by the Basel Committee and some banks themselves have a target for the capital they will hold that is higher than that specified by their bank supervisors.

25.4 The G-30 Policy Recommendations

In 1993, a working group consisting of end-users, dealers, academics, accountants, and lawyers involved in derivatives published a report that contained 20 risk management recommendations for dealers and end-users of derivatives and four recommendations for legislators, regulators, and supervisors. The report was based on a detailed survey of 80 dealers and 72 end-users worldwide. The survey involved both questionnaires and in-depth interviews. The report is not a regulatory document, but it has been influential in the development of risk management practices.

A brief summary of the important recommendations is as follows:

1. A company's policies on risk management should be clearly defined and approved by senior management, ideally at the board of directors level. Managers at all levels should enforce the policies.

[3]Noncumulative perpetual preferred stock is preferred stock lasting forever where there is a predetermined dividend rate. Unpaid dividends do not cumulate (that is, the dividends for one year are not carried forward to the next year).

[4]Goodwill is created when one company acquires another. It is the purchase price minus the book value of the assets. It represents the intangible assets acquired.

[5]In cumulative preferred stock, unpaid dividends cumulate. Any backlog of dividends must be paid before dividends are paid on the common stock.

2. Derivatives positions should be marked to market (i.e., revalued using a model that is consistent with market prices) at least once a day.
3. Derivatives dealers should measure market risk using a consistent measure such as value at risk. Limits to the market risks that are taken should be set.
4. Derivatives dealers should carry out stress tests to determine potential losses under extreme market conditions.
5. The risk management function should be set up so that it is independent of the trading operation.
6. Credit exposures arising from derivatives trading should be assessed based on the current replacement value of existing positions and potential future replacement costs.
7. Credit exposures to a counterparty should be aggregated in a way that reflects enforceable netting agreements. (We talk about netting in the next section.)
8. The individuals responsible for setting credit limits should be independent of those involved in trading.
9. Dealers and end-users should assess carefully both the costs and benefits of credit risk mitigation techniques such as collateralization and downgrade triggers. In particular, they should assess their own capacity and that of their counterparties to meet the cash flow requirement of downgrade triggers. (Downgrade triggers are discussed in Chapter 18.)
10. Only individuals with the appropriate skills and experience should be allowed to have responsibility for trading derivatives, supervising the trading, carrying out back office functions in relation to the trading, and so on.
11. There should be adequate systems in place for data capture, processing, settlement, and management reporting.
12. Dealers and end-users should account for the derivatives transactions used to manage risks so as to achieve a consistency of income recognition treatment between those instruments and the risks being managed.

25.5 Netting

Participants in the over-the-counter derivatives market have traditionally signed an International Swaps and Derivatives Association (ISDA) master agreement covering their derivatives trades. The word *netting* refers to a clause in the master agreement, which states that in the event of a default all transactions are considered as a single transaction. Effectively, this means that, if a company defaults on one transaction that is covered by the master agreement, it must default on all transactions covered by the master agreement.

Netting and ISDA master agreements were discussed in Chapters 6 and 18. Netting can have the effect of substantially reducing credit risk. Consider a bank that has three swap transactions outstanding with a particular counterparty. The transactions are worth +\$24 million, –\$17 million, and +\$8 million to the bank. Suppose that the counterparty experiences financial difficulties and defaults on its outstanding obligations.

To the counterparty the three transactions have values of −\$24 million, +\$17 million, and −\$8 million, respectively. Without netting, the counterparty would default on the first transaction, keep the second transaction, and default on the third transaction. Assuming no recovery, the loss to the bank would be \$32 (= 24 + 8) million. With netting, the counterparty is required to default on the second transaction as well. The loss to the bank is then \$15 (= 24 − 17 + 8) million.

More generally, suppose that a financial institution has a portfolio of N derivatives outstanding with a particular counterparty and that the current value of the ith derivative is V_i. Without netting, the financial institution's exposure in the event of a default today is

$$\sum_{i=1}^{N} \max(V_i, 0)$$

With netting, it is

$$\max\left(\sum_{i=1}^{N} V_i, 0\right)$$

Without netting, the exposure is the payoff from a portfolio of options. With netting, the exposure is the payoff from an option on a portfolio.

The 1988 Basel Accord did not take netting into account in setting capital requirements. From equation (25.1) the credit equivalent amount for a portfolio of derivatives with a counterparty under the Accord was

$$\sum_{i=1}^{N} [\max(V_i, 0) + a_i L_i]$$

where a_i is the add-on factor for the ith transaction and L_i is the principal for the ith transaction.

By 1995, netting had been successfully tested in the courts in many jurisdictions. As a result, the 1988 Accord was modified to allow banks to reduce their credit equivalent totals when enforceable bilateral netting agreements were in place. The first step was to calculate the net replacement ratio, NRR. This is the ratio of the current exposure with netting to the current exposure without netting:

$$\text{NRR} = \frac{\max(\sum_{i=1}^{N} V_i, 0)}{\sum_{i=1}^{N} \max(V_i, 0)}$$

The credit equivalent amount was modified to

$$\max\left(\sum_{i=1}^{N} V_i, 0\right) + (0.4 + 0.6 \times \text{NRR}) \sum_{i=1}^{N} a_i L_i$$

Table 25.3 Portfolio of Derivatives with a Particular Counterparty

Transaction	**Principal, L_i**	**Current Value, V_i**	**Add-On Amount, a_iL_i**
3-year interest rate swap	1,000	−60	5
6-year foreign exchange forward	1,000	70	75
9-month option on a stock	500	55	30

Example 25.4

Consider the example in Table 25.3, which shows a portfolio of three derivatives that a bank has with a particular counterparty. The third column shows the current mark-to-market values of the transactions and the fourth column shows the add-on amount calculated from Table 25.2. The current exposure with netting is $-60 + 70 + 55 = 65$. The current exposure without netting is $0 + 70 + 55 = 125$. The net replacement ratio is given by

$$\text{NRR} = \frac{65}{125} = 0.52$$

The total of the add-on amounts, $\sum a_i L_i$, is $5 + 75 + 30 = 110$. The credit equivalent amount when netting agreements are in place is $65 + (0.4 + 0.6 \times 0.52) \times 110 = 143.32$. Without netting, the credit equivalent amount is $125 + 110 = 235$. Suppose that the counterparty is an OECD bank so that the risk weight is 0.2. This means that the risk-weighted assets with netting is $0.2 \times 143.32 = 28.66$. Without netting, it is $0.2 \times 235 = 47$.

25.6 The 1996 Amendment

In 1995, the Basel Committee issued a consultative proposal to amend the 1988 Accord. This became known as the "1996 Amendment." It was implemented in 1998 and was then sometimes referred to as "BIS 98." The amendment involves keeping capital for the market risks associated with trading activities.

Marking to market is the practice of revaluing assets and liabilities daily using a model that is calibrated to current market prices. It is also known as *fair value accounting*. Banks are required to use fair value accounting for all assets and liabilities that are held for trading purposes. This includes most derivatives, marketable equity securities, foreign currencies, and commodities. These items constitute what is referred to as the bank's *trading book*.

Under the 1996 Amendment, the credit risk capital charge in the 1988 Accord continued to apply to all on-balance-sheet and off-balance-sheet items in the trading and banking book, except positions in the trading book that consisted of (a) debt and equity traded securities and (b) positions in commodities and foreign exchange. The

Amendment introduced a capital charge for the market risk associated with all items in the trading book.[6]

The 1996 Amendment outlined a standardized approach for measuring the capital charge for market risk. The standardized approach assigned capital separately to each of debt securities, equity securities, foreign exchange risk, commodities risk, and options. No account was taken of correlations between different types of instruments. The more sophisticated banks with well-established risk management functions were allowed to use an "internal model-based approach" for setting market risk capital. This involved calculating a value-at-risk measure and converting it into a capital requirement using a formula specified in the 1996 Amendment. Most large banks preferred to use the internal model-based approach because it better reflected the benefits of diversification and led to lower capital requirements.

The value-at-risk measure used in the internal model-based approach was calculated with a 10-day time horizon and a 99% confidence level. It is the loss that has a 1% chance of being exceeded over a 10-day period. The capital requirement is

$$\max(\mathrm{VaR}_{t-1}, m_c \times \mathrm{VaR}_{\mathrm{avg}}) + \mathrm{SRC} \tag{25.3}$$

where m_c is a multiplicative factor, and SRC is a *specific risk charge*. The variable VaR_{t-1} is the previous day's value at risk and $\mathrm{VaR}_{\mathrm{avg}}$ is the average value at risk over the past 60 days. The minimum value for m_c is 3. Higher values may be chosen by regulators for a particular bank if tests reveal inadequacies in the bank's value-at-risk model, as will be explained shortly.

The first term in equation (25.3) covers risks relating to movements in broad market variables such as interest rates, exchange rates, stock indices, and commodity prices. The second term, SRC, covers risks related to specific companies such as those concerned with movements in a company's stock price or changes in a company's credit spread.

Consider the first term and assume $m_c = 3$. In most circumstances, the most recently calculated VaR, VaR_{t-1}, is less than three times the average VaR over the past 60 days. This gives the capital requirement for movements in broad market variables as simply

$$3 \times \mathrm{VaR}_{\mathrm{avg}}$$

The most popular method for calculating VaR is historical simulation, described in Chapter 12. As explained in Chapter 11, banks almost invariably calculate a one-day 99% VaR in the first instance. When formulating the 1996 amendment, regulators explicitly stated that the 10-day 99% VaR can be calculated as $\sqrt{10}$ times the one-day 99% VaR. This means that, when the capital requirement for a bank is calculated as m_c times the average 10-day 99% VaR, it is to all intents and purposes $m_c \times \sqrt{10} = 3.16m_c$ times the average one-day 99% VaR. If $m_c = 3$, it is 9.48 times the average one-day VaR.

Consider next SRC. One security that gives rise to an SRC is a corporate bond. There are two components to the risk of this security: interest rate risk and credit risk of

[6]Certain nontrading book positions that are used to hedge positions in the trading book can be included in the calculation of the market risk capital charge.

the corporation issuing the bond. The interest rate risk is captured by the first term in equation (25.3); the credit risk is captured by the SRC.[7] The 1996 Amendment proposed standardized methods for assessing the SRC, but allowed banks to use internal models once regulatory approval for the models had been obtained.

The internal model for SRC must involve calculating a 10-day 99% value at risk for specific risks. Regulators calculate capital by applying a multiplicative factor (similar to m_c) to the value at risk. This multiplicative factor must be at least 4 and the resultant capital must be at least 50% of the capital given by the standardized approach. An approach for calculating SRC for credit-spread risk was indicated in Section 19.5.

The total capital a bank was required to keep after the implementation of the 1996 Amendment was the sum of (a) credit risk capital equal to 8% of the risk-weighted assets (RWA) and (b) market risk capital as explained in this section. For convenience, an RWA for market risk capital was defined as 12.5 multiplied by the amount given in equation (25.3). This means that the total capital required for credit and market risk is given by

$$\text{Total Capital} = 0.08 \times (\text{credit risk RWA} + \text{market risk RWA}) \tag{25.4}$$

A bank had more flexibility in the type of capital it used for market risk. It could use Tier 1 or Tier 2 capital. It could also use what is termed Tier 3 capital. This consists of short-term subordinated debt with an original maturity of at least two years that is unsecured and fully paid up. (Tier 3 capital was eliminated under Basel III.)

25.6.1 Back-Testing

The BIS Amendment required the one-day 99% VaR that a bank calculates to be back-tested over the previous 250 days. As described in Section 11.10, this involves using the bank's current procedure for estimating VaR for each of the most recent 250 days. If the actual loss that occurred on a day is greater than the VaR level calculated for the day, an "exception" is recorded. Calculations are carried out (a) including changes that were made to the portfolio on the day being considered and (b) assuming that no changes were made to the portfolio on the day being considered. (Regulators like to see both calculations.)

If the number of exceptions during the previous 250 days is less than 5, m_c is normally set equal to 3. If the number of exceptions is 5, 6, 7, 8, and 9, the value of the m_c is set equal to 3.4, 3.5, 3.65, 3.75, and 3.85, respectively. The bank supervisor has some discretion as to whether the higher multipliers are used. They will normally be applied when the reason for the exceptions is identified as a deficiency in the VaR model being used. If changes in the bank's positions during the day result in exceptions, the higher multiplier should be considered, but does not have to be used. When the only reason that is identified is bad luck, no guidance is provided for the supervisor. In circumstances

[7] As mentioned earlier, the 1988 credit risk capital charge did not apply to debt securities in the trading book under the 1996 Amendment.

where the number of exceptions is 10 or more, the Basel Amendment requires the multiplier to be set at 4. Problem 25.18 considers these guidelines in the context of the statistical tests we discussed in Chapter 11.

25.7 Basel II

The 1988 Basel Accord improved the way capital requirements were determined, but it does have significant weaknesses. Under the Accord, all loans by a bank to a corporation have a risk weight of 100% and require the same amount of capital. A loan to a corporation with a AAA credit rating is treated in the same way as one to a corporation with a B credit rating.[8] Also, in Basel I there was no model of default correlation.

In June 1999, the Basel Committee proposed new rules that have become known as Basel II. These were revised in January 2001 and April 2003. A number of Quantitative Impact Studies (QISs) were carried out prior to the implementation of the new rules to test them by calculating the amount of capital that would be required if the rules had been in place.[9] A final set of rules agreed to by all members of the Basel Committee was published in June 2004. This was updated in November 2005. Implementation of the rules began in 2007 after a further QIS.

The Basel II capital requirements applied to “internationally active” banks. In the United States, there are many small regional banks and the U.S. regulatory authorities decided that Basel II would not apply to them. (These banks are regulated under what is termed Basel IA, which is similar to Basel I.) In Europe, all banks, large or small, were regulated under Basel II. Furthermore, the European Union required the Basel II rules to be applied to securities companies as well as banks.

The Basel II is based on three “pillars”:

1. Minimum Capital Requirements
2. Supervisory Review
3. Market Discipline

In Pillar 1, the minimum capital requirement for credit risk in the banking book is calculated in a new way that reflects the credit risk of counterparties. The capital requirement for market risk remains unchanged from the 1996 Amendment and there is a new capital charge for operational risk. The general requirement in Basel I that banks hold a total capital equal to 8% of risk-weighted assets (RWA) remains unchanged. When the capital requirement for a particular risk is calculated directly rather than in a way involving RWAs, it is multiplied by 12.5 to convert it into an RWA-equivalent. As a result it is always the case that

[8]Credit ratings are discussed in Section 1.7.

[9]One point to note about the QISs is that they do not take account of changes banks may choose to make to their portfolios to minimize their capital requirements once the new rules have been implemented.

$$\text{Total Capital} = 0.08 \times (\text{credit risk RWA} + \text{market risk RWA} + \text{operational risk RWA}) \tag{25.5}$$

Pillar 2 is concerned with the supervisory review process. It covers both quantitative and qualitative aspects of the ways risk is managed within a bank. Supervisors are required to ensure that a bank has a process in place for ensuring that capital levels are maintained. Banks are expected to keep more than the minimum regulatory capital to allow for fluctuations in capital requirements and possible difficulties in raising capital at short notice. Regulators in different countries are allowed some discretion in how rules are applied (so that they can take account of local conditions), but overall consistency in the application of the rules is required. Pillar 2 places more emphasis on early intervention when problems arise. Supervisors are required to do far more than just ensuring that the minimum capital required under Basel II is held. Part of their role is to encourage banks to develop and use better risk management techniques and to evaluate those techniques. They should evaluate risks that are not covered by Pillar 1 (e.g., concentration risks) and enter into an active dialogue with banks when deficiencies are identified.

The third pillar, market discipline, requires banks to disclose more information about the way they allocate capital and the risks they take. The idea here is that banks will be subjected to added pressure to make sound risk management decisions if shareholders and potential shareholders have more information about those decisions.

25.8 Credit Risk Capital Under Basel II

For credit risk, Basel II specified three approaches:

1. The Standardized Approach
2. The Foundation Internal Ratings Based (IRB) Approach
3. The Advanced IRB Approach

However, the United States (which, as mentioned earlier, chose to apply Basel II only to large banks) decided that only the IRB approach can be used.

25.8.1 The Standardized Approach

The standardized approach is used by banks that are not sufficiently sophisticated (in the eyes of the regulators) to use the internal ratings approaches. The standardized approach is similar to Basel I except for the calculation of risk weights.[10] Some of the

[10]Ratios calculated using the new weights are sometimes referred to as McDonough ratios after William McDonough, the head of the Basel Committee.

Table 25.4 Risk Weights as a Percent of Principal for Exposures to Countries, Banks, and Corporations Under Basel II's Standardized Approach

	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to BB–	B+ to B–	Below B–	Unrated
Country*	0	20	50	100	100	150	100
Banks**	20	50	50	100	100	150	50
Corporations	20	50	100	100	150	150	100

*Includes exposures to the country's central bank.
**National supervisors have options as outlined in the text.

new rules here are summarized in Table 25.4. Comparing Table 25.4 with Table 25.1, we see that the OECD status of a bank or a country is no longer considered important under Basel II. The risk weight for a country (sovereign) exposure ranges from 0% to 150% and the risk weight for an exposure to another bank or a corporation ranges from 20% to 150%. In Table 25.1, OECD banks were implicitly assumed to be lesser credit risks than corporations. An OECD bank attracted a risk weight of 20% while a corporation attracted a risk weight of 100%. Table 25.4 treats banks and corporations much more equitably. An interesting observation from Table 25.4 for a country, corporation, or bank that wants to borrow money is that it may be better to have no credit rating at all than a very poor credit rating. (Usually a company gets a credit rating when it issues a publicly traded bond.) Supervisors are allowed to apply lower risk weights (20% rather than 50%, 50% rather than 100%, and 100% rather than 150%) when exposures are to the country in which the bank is incorporated or to that country's central bank.

The risk weight for mortgages in the Basel II standardized approach was 35% and that for other retail loans was 75%. For claims on banks, the rules are somewhat complicated. Instead of using the risk weights in Table 25.4, national supervisors can choose to base capital requirements on the rating of the country in which the bank is incorporated. The risk weight assigned to the bank will be 20% if the country of incorporation has a rating between AAA and AA–, 50% if it is between A+ and A–, 100% if it is between BBB+ and B–, 150% if it is below B–, and 100% if it is unrated. Another complication is that, if national supervisors elect to use the rules in Table 25.4, they can choose to treat claims with a maturity less than three months more favorably so that the risk weights are 20% if the rating is between AAA+ and BBB–, 50% if it is between BB+ and B–, 150% if it is below B–, and 20% if it is unrated.

In December 2017, the Basel Committee created a revised, more granular standardized approach for determining risk weights for credit exposures. It also limited the use of the Advanced IRB approaches in some situations. As explained in Chapter 26, the revised standardized approach provides a floor for the determination of total capital requirements.

Example 25.5

Suppose that the assets of a bank consist of $100 million of loans to corporations rated A, $10 million of government bonds rated AAA, and $50 million of residential mortgages. Under the Basel II standardized approach, the total of the risk-weighted assets is

$$0.5 \times 100 + 0.0 \times 10 + 0.35 \times 50 = 67.5$$

or $67.5 million. This compares with $125 million under Basel I. (See Example 25.1.)

25.8.2 Adjustments for Collateral

There are two ways banks can adjust risk weights for collateral under Basel II. The first is termed the *simple approach* and is similar to an approach used in Basel I. The second is termed the *comprehensive approach*. Banks have a choice as to which approach is used in the banking book, but must use the comprehensive approach to calculate capital for counterparty credit risk in the trading book.

Under the simple approach, the risk weight of the counterparty is replaced by the risk weight of the collateral for the part of the exposure covered by the collateral. (The exposure is calculated after netting.) For any exposure not covered by the collateral, the risk weight of the counterparty is used. The minimum level for the risk weight applied to the collateral is 20%.[11] A requirement is that the collateral must be revalued at least every six months and must be pledged for at least the life of the exposure.

Under the comprehensive approach, banks adjust the size of their exposure upward to allow for possible increases in the exposure and adjust the value of the collateral downward to allow for possible decreases in the value of the collateral.[12] (The adjustments depend on the volatility of the exposure and the collateral.) A new exposure equal to the excess of the adjusted exposure over the adjusted value of the collateral is calculated and the counterparty's risk weight is applied to this exposure. The adjustments applied to the exposure and the collateral can be calculated using rules specified in Basel II or, with regulatory approval, using a bank's internal models. Where netting arrangements apply, exposures and collateral are separately netted and the adjustments made are weighted averages.

Example 25.6

Suppose that an $80 million exposure to a particular counterparty is secured by collateral worth $70 million. The collateral consists of bonds issued by an A-rated company. The

[11]An exception is when the collateral consists of cash or government securities with the currency of the collateral being the same as the currency of the exposure.

[12]An adjustment to the exposure is not likely to be necessary on a loan, but is likely to be necessary on an over-the-counter derivative. The adjustment is in addition to the add-on factor.

counterparty has a rating of B+. The risk weight for the counterparty is 150% and the risk weight for the collateral is 50%. The risk-weighted assets applicable to the exposure using the simple approach is

$$0.5 \times 70 + 1.50 \times 10 = 50$$

or \$50 million.

Consider next the comprehensive approach. Assume that the adjustment to exposure to allow for possible future increases in the exposure is +10% and the adjustment to the collateral to allow for possible future decreases in its value is –15%. The new exposure is

$$1.1 \times 80 - 0.85 \times 70 = 28.5$$

or \$28.5 million, and a risk weight of 150% is applied to this exposure to give risk-adjusted assets equal to \$42.75 million.

25.8.3 The IRB Approach

The model underlying the IRB approach is shown in Figure 25.1. Regulators base the capital requirement on the value at risk calculated using a one-year time horizon and a 99.9% confidence level. They recognize that expected losses are usually covered by the way a financial institution prices its products. (For example, the interest charged by a bank on a loan portfolio is designed to recover expected loan losses on the portfolio.) The capital required is therefore the value at risk minus the expected loss.

The value at risk is calculated using the one-factor Gaussian copula model of time to default that we discussed in Section 9.6. Assume that a bank has a very large number of obligors and the *i*th obligor has a one-year probability of default equal to PD_i. The copula correlation between each pair of obligors is ρ.[13] As in Section 9.6, we define

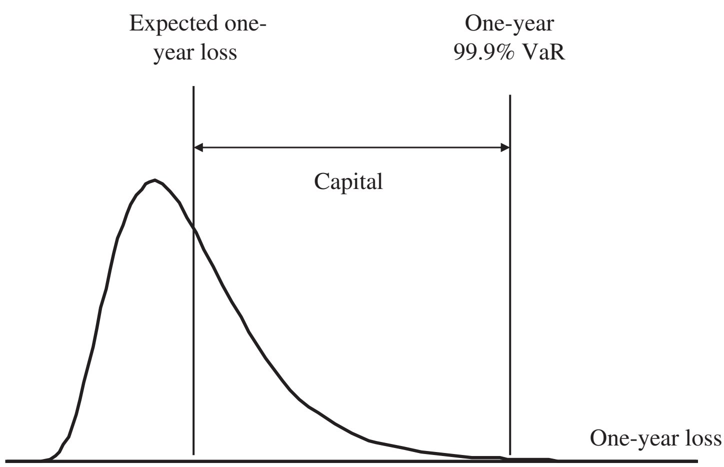

Figure 25.1 The Loss Probability Density Function and the Capital Required by a Financial Institution

[13]Note that the Basel Committee publications use R, not ρ, to denote the copula correlation.

$$\mathrm{WCDR}_i = N\left[\frac{N^{-1}(\mathrm{PD}_i) + \sqrt{\rho}N^{-1}(0.999)}{\sqrt{1-\rho}}\right] \tag{25.6}$$

where WCDR_i denotes the "worst-case default rate" defined so that the bank is 99.9% certain it will not be exceeded next year for the ith counterparty. Gordy's (2003) research shows that for a large portfolio of instruments (loans, loan commitments, derivatives, and so on) that have the same ρ, in a one-factor model the one-year 99.9% VaR is approximately[14]

$$\sum_i \mathrm{EAD}_i \times \mathrm{LGD}_i \times \mathrm{WCDR}_i$$

where EAD_i is the exposure at default of the ith counterparty and LGD_i is the loss given default for the ith counterparty.

The variable EAD_i is the dollar amount that is expected to be owed by the ith counterparty at the time of default during the next year. The variable LGD_i is the proportion of EAD_i that is expected to be lost in the event of default. For example, if a bank expects to recover 30% of the amount owed in the event of default, $\mathrm{LGD}_i = 0.7$.

The expected loss from defaults is

$$\sum_i \mathrm{EAD}_i \times \mathrm{LGD}_i \times \mathrm{PD}_i$$

The capital required in Figure 25.1 is the excess of the 99.9% worst-case loss over the expected loss. It is therefore

$$\sum_i \mathrm{EAD}_i \times \mathrm{LGD}_i \times (\mathrm{WCDR}_i - \mathrm{PD}_i) \tag{25.7}$$

We now drop the subscripts and define for a counterparty:

PD: The probability that the counterparty will default within one year (expressed as a decimal)
EAD: The exposure at default (in dollars)
LGD: The loss given default or the proportion of the exposure that is lost if there is a default (expressed as a decimal)

Table 25.5 shows how WCDR depends on PD and ρ in the Gaussian copula model. When the correlation ρ is zero, WCDR = PD because in that case there is no default correlation and the percentage of loans defaulting can be expected to be the same in all years. As ρ increases, WCDR increases.

[14]See M. B. Gordy, "A Risk-Factor Model Foundation for Ratings-Based Bank Capital Ratios," *Journal of Financial Intermediation* 12 (2003): 199–232.

Table 25.5 Dependence of One-Year 99.9% WCDR on PD and ρ

	PD = 0.1%	PD = 0.5%	PD = 1%	PD = 1.5%	PD = 2.0%
$\rho = 0.0$	0.1%	0.5%	1.0%	1.5%	2.0%
$\rho = 0.2$	2.8%	9.1%	14.6%	18.9%	22.6%
$\rho = 0.4$	7.1%	21.1%	31.6%	39.0%	44.9%
$\rho = 0.6$	13.5%	38.7%	54.2%	63.8%	70.5%
$\rho = 0.8$	23.3%	66.3%	83.6%	90.8%	94.4%

25.8.4 Exposure at Default

In the Foundation IRB approach, the calculation of EAD for derivatives is usually based on the CEM approach of Basel I.[15] In the Advanced IRB approach, banks can use their internal models to calculate EAD.

The first stage when internal models are used is to calculate expected exposure (EE) to each counterparty at a number of future times. (This generally involves Monte Carlo simulation.) The Effective Expected Exposure (Effective EE) at a future time, t, is the maximum EE between time zero an time t. EAD is set equal to 1.4 times the Effective Expected Positive Exposure, which is the average of the Effective EEs over the first year.[16]

25.8.5 Corporate, Sovereign, and Bank Exposures

In the case of corporate, sovereign, and bank exposures, Basel II assumes a relationship between the correlation parameter, ρ, and the probability of default, PD, based on empirical research.[17] The formula is

$$\rho = 0.12\frac{1 - \exp(-50 \times \text{PD})}{1 - \exp(-50)} + 0.24\left[1 - \frac{1 - \exp(-50 \times \text{PD})}{1 - \exp(-50)}\right]$$

Because $\exp(-50)$ is a very small number, this formula is to all intents and purposes

$$\rho = 0.12(1 + e^{-50\times\text{PD}}) \tag{25.8}$$

As PD increases, ρ decreases. The reason usually given for this inverse relationship is as follows. As a company becomes less creditworthy, its PD increases and its probability of default becomes more idiosyncratic and less affected by overall market conditions.

Combining equation (25.8) with equation (25.6), we obtain the relationship between WCDR and PD in Table 25.6. WCDR is, as one would expect, an increasing function of

[15] As will be discussed in Section 26.4, this was revised under a regulatory change known as SA-CCR.
[16] See Basel Committee on Banking Supervision, "The Application of Basel II to Trading Activities and the Treatment of Double Default Effects," July 2005.
[17] See J. Lopez, "The Empirical Relationship between Average Asset Correlation, Firm Probability of Default, and Asset Size," *Journal of Financial Intermediation* 13, no. 2 (2004): 265–283.

Table 25.6 Relationship between One-Year 99.9% WCDR and PD for Corporate, Sovereign, and Bank Exposures

PD	0.1%	0.5%	1%	1.5%	2.0%
WCDR	3.4%	9.8%	14.0%	16.9%	19.0%

PD. However, it does not increase as fast as it would if ρ were assumed to be independent of PD.

The formula for the capital required for the counterparty is

$$\text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA} \tag{25.9}$$

The meaning of the first three terms in this expression should be clear from our earlier discussion leading to equation (25.7). The variable MA is the maturity adjustment and is defined as

$$\text{MA} = \frac{1 + (\text{M} - 2.5) \times b}{1 - 1.5 \times b} \tag{25.10}$$

where

$$b = [0.11852 - 0.05478 \times \ln(\text{PD})]^2$$

and M is the maturity of the exposure. The maturity adjustment is designed to allow for the fact that, if an instrument lasts longer than one year, there is a one-year credit exposure arising from a possible decline in the creditworthiness of the counterparty as well as from a possible default by the counterparty. (Note that, when $\text{M} = 1$, MA is 1.0 and has no effect.) As mentioned in Section 25.7 (see Equation (25.5)), the risk-weighted assets (RWA) are calculated as 12.5 times the capital required

$$\text{RWA} = 12.5 \times \text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA}$$

so that the capital is 8% of RWA, 4% of which must be Tier 1.

Under the Foundation IRB approach, banks supply PD while LGD, EAD, and M are supervisory values set by the Basel Committee. PD is subject to a floor of 0.03% for bank and corporate exposures. LGD is set at 45% for senior claims and 75% for subordinated claims. When there is eligible collateral, in order to correspond to the comprehensive approach that we described earlier, LGD is reduced by the ratio of the adjusted value of the collateral to the adjusted value of the exposure, both calculated using the comprehensive approach. For derivatives, the EAD is calculated using the CEM "current exposure plus add-on" approach of Basel I and includes the impact of netting. M is set at 2.5 in most circumstances.

Under the Advanced IRB approach, banks supply their own estimates of the PD, LGD, EAD, and M for corporate, sovereign, and bank exposures. The PD can be reduced by credit mitigants such as credit triggers. (As in the case of the Foundation IRB approach, it is subject to a floor of 0.03% for bank and corporate exposures.) The two main factors influencing the LGD are the seniority of the debt and the collateral.

In calculating EAD, banks can with regulatory approval use their own models. As mentioned, in the case of derivatives, the model is likely to involve Monte Carlo simulation to determine how expected exposure (after netting and collateral) will vary over the next year.

The capital given by equation (25.9) is intended to be sufficient to cover unexpected losses over a one-year period that we are 99.9% certain will not be exceeded. (As discussed earlier, the expected losses should be covered by a bank in the way it prices its products.) The WCDR is the default rate that (theoretically) happens once every thousand years. The Basel Committee reserved the right to apply a scaling factor (less than or greater than 1.0) to the result of the calculations in equation (25.9) if it finds that the aggregate capital requirements are too high or low. A typical scaling factor is 1.06.

Example 25.7

Suppose that the assets of a bank consist of \$100 million of loans to A-rated corporations. The PD for the corporations is estimated as 0.1% and the LGD is 60%. The average maturity is 2.5 years for the corporate loans. This means that

$$b = [0.11852 - 0.05478 \times \ln(0.001)]^2 = 0.247$$

so that

$$\mathrm{MA} = \frac{1}{1 - 1.5 \times 0.247} = 1.59$$

From Table 25.6, the WCDR is 3.4%. Under the Basel II IRB approach, the risk-weighted assets for the corporate loans are

$$12.5 \times 100 \times 0.6 \times (0.034 - 0.001) \times 1.59 = 39.3$$

or \$39.3 million. This compares with \$100 million under Basel I and \$50 million under the standardized approach of Basel II. (See Examples 25.1 and 25.5 where a \$100 million corporate loan is part of the portfolio.)

25.8.6 Retail Exposures

The model underlying the calculation of capital for retail exposures in Basel II is similar to that underlying the calculation of corporate, sovereign, and banking exposures. However, the Foundation IRB and Advanced IRB approaches are merged and all banks using the IRB approach provide their own estimates of PD, EAD, and LGD. There is no maturity adjustment, MA. The capital requirement is therefore

$$\mathrm{EAD} \times \mathrm{LGD} \times (\mathrm{WCDR} - \mathrm{PD})$$

and the risk-weighted assets are

$$12.5 \times \mathrm{EAD} \times \mathrm{LGD} \times (\mathrm{WCDR} - \mathrm{PD})$$

Table 25.7 Relationship between One-Year 99.9% WCDR and PD for Retail Exposures

PD	0.1%	0.5%	1.0%	1.5%	2.0%
WCDR	2.1%	6.3%	9.1%	11.0%	12.3%

WCDR is calculated as in equation (25.6). For residential mortgages, ρ is set equal to 0.15 in this equation.[18] For qualifying revolving exposures, ρ is set equal to 0.04. For all other retail exposures, a relationship between ρ and PD is specified for the calculation of WCDR. This is

$$\rho = 0.03\frac{1 - \exp(-35 \times \text{PD})}{1 - \exp(-35)} + 0.16\left[1 - \frac{1 - \exp(-35 \times \text{PD})}{1 - \exp(-35)}\right]$$

Because $\exp(-35)$ is a very small number, this formula is to all intents and purposes

$$\rho = 0.03 + 0.13e^{-35\times\text{PD}} \tag{25.11}$$

Comparing equation (25.11) with equation (25.8), we see that correlations are assumed to be much lower for retail exposures than for corporate exposures. Table 25.7 is the table corresponding to Table 25.6 for retail exposures.

Example 25.8

Suppose that the assets of a bank consist of \$50 million of residential mortgages where the PD is 0.005 and the LGD is 20%. In this case, $\rho = 0.15$ and

$$\text{WCDR} = N\left[\frac{N^{-1}(0.005) + \sqrt{0.15}N^{-1}(0.999)}{\sqrt{1 - 0.15}}\right] = 0.067$$

The risk-weighted assets are

$$12.5 \times 50 \times 0.2 \times (0.067 - 0.005) = 7.8$$

or \$7.8 million. This compares with \$25 million under Basel I and \$17.5 million under the Standardized Approach of Basel II. (See Examples 25.1 and 25.5 where \$50 million of residential mortgages is part of the portfolio.)

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[18]In the light of experience during the crisis, this is probably too low.

25.8.7 Guarantees and Credit Derivatives

The approach traditionally taken by the Basel Committee for handling guarantees and credit derivatives such as credit default swaps is the credit substitution approach. Suppose that a AA-rated company guarantees a loan to a BBB-rated company. For the purposes of calculating capital, the credit rating of the guarantor is substituted for the credit rating of the borrower so that capital is calculated as though the loan had been made to the AA-rated company. This overstates the credit risk because, for the lender to lose money, both the guarantor and the borrower must default (with the guarantor defaulting before the borrower).[19] The Basel Committee has addressed this issue. In July 2005, it published a document concerned with the treatment of double defaults under Basel II.[20] As an alternative to using the credit substitution approach, the capital requirement can be calculated as the capital that would be required without the guarantee multiplied by $0.15 + 160 \times \text{PD}_g$ where PD_g is the one-year probability of default of the guarantor.

25.9 Operational Risk Capital Under Basel II

In addition to changing the way banks calculate credit risk capital, Basel II requires banks to keep capital for operational risk. This is the risk of losses from situations where the bank's procedures fail to work as they are supposed to or where there is an adverse external event such as a fire in a key facility.

There were three approaches to calculating capital for operational risk in Basel II:

1. The Basic Indicator Approach
2. The Standardized Approach
3. The Advanced Measurement Approach

Which of these was used depended on the sophistication of the bank. The simplest approach is the Basic Indicator Approach. This sets the operational risk capital equal to the bank's average annual gross income over the last three years multiplied by 0.15.[21] The Standardized Approach is similar to the basic indicator approach except that a different factor is applied to the gross income from different business lines. In the Advanced Measurement Approach, the bank uses its own internal models to calculate the operational risk loss that it is 99.9% certain will not be exceeded in one year. Similarly to the way credit risk capital is calculated in the IRB approach, operational risk capital

[19]Credit default swaps, which we discussed in Chapter 17, provide a type of insurance against default and are handled similarly to guarantees for regulatory purposes.

[20]See Bank for International Settlements, "The Application of Basel II to Trading Activities and the Treatment of Double Defaults," July 2005, available on www.bis.org.

[21]Gross income defined as net interest income plus non-interest income. Net interest income is the excess of income earned on loans over interest paid on deposits and other instruments that are used to fund the loans. Years where gross income is negative are not included in the calculations.

was set equal to this loss minus the expected loss. One advantage of the advanced measurement approach was that it allowed banks to recognize the risk-mitigating impact of insurance contracts subject to certain conditions. However, because there was very little consistency between the calculations of different banks, it has now been replaced by the standardized measurement approach described in Section 20.4.

25.10 Pillar 2: Supervisory Review

Pillar 2 of Basel II is concerned with the supervisory review process. Four key principles of supervisory review were specified:

1. Banks should have a process for assessing their overall capital adequacy in relation to their risk profile and a strategy for maintaining their capital levels.
2. Supervisors should review and evaluate banks' internal capital adequacy assessments and strategies, as well as their ability to monitor and ensure compliance with regulatory capital ratios. Supervisors should take appropriate supervisory action if they are not satisfied with the result of this process.
3. Supervisors should expect banks to operate above the minimum regulatory capital and should have the ability to require banks to hold capital in excess of this minimum.
4. Supervisors should seek to intervene at an early stage to prevent capital from falling below the minimum levels required to support the risk characteristics of a particular bank and should require rapid remedial action if capital is not maintained or restored.

The Basel Committee suggested that regulators pay particular attention to interest rate risk in the banking book, credit risk, and operational risk. Key issues in credit risk are stress tests used, default definitions used, credit risk concentration, and the risks associated with the use of collateral, guarantees, and credit derivatives.

The Basel Committee also stressed that there should be transparency and accountability in the procedures used by bank supervisors. This is particularly important when a supervisor exercises discretion in the procedures used or sets capital requirements above the minimum specified in Basel II.

25.11 Pillar 3: Market Discipline

Pillar 3 of Basel II is concerned with increasing the disclosure by a bank of its risk assessment procedures and capital adequacy. The extent to which regulators can force banks to increase their disclosure varies from jurisdiction to jurisdiction. However, banks are unlikely to ignore directives on this from their supervisors, given the potential of supervisors to make their life difficult. Also, in some instances, banks have to increase their disclosure in order to be allowed to use particular methodologies for calculating capital.

Regulatory disclosures are likely to be different in form from accounting disclosures and need not be made in annual reports. It is largely left to the bank to choose disclosures that are material and relevant. Among the items that banks should disclose are:

1. The entities in the banking group to which Basel II is applied and adjustments made for entities to which it is not applied
2. The terms and conditions of the main features of all capital instruments
3. A list of the instruments constituting Tier 1 capital and the amount of capital provided by each item
4. The total amount of Tier 2 capital
5. Capital requirements for credit, market, and operational risk
6. Other general information on the risks to which a bank is exposed and the assessment methods used by the bank for different categories of risk
7. The structure of the risk management function and how it operates

25.12 Solvency II

As discussed in Section 3.11, there are no international standards for the regulation of insurance companies. In the United States, insurance companies are regulated at the state level with some input from the Federal Insurance Office and the National Association of Insurance Commissioners. In Europe, the regulation of insurance companies is handled by the European Union. The long-standing regulatory framework in the European Union, known as Solvency I, was replaced by Solvency II in 2016. Whereas Solvency I calculates capital only for underwriting risks, Solvency II considers investment risks and operational risks as well.

Solvency II has many similarities to Basel II. There are three pillars. Pillar 1 is concerned with the calculation of capital requirements and the types of capital that are eligible. Pillar 2 is concerned with the supervisory review process. Pillar 3 is concerned with the disclosure of risk management information to the market. The three pillars are therefore analogous to the three pillars of Basel II.

Pillar 1 of Solvency II specifies a *minimum capital requirement* (MCR) and a *solvency capital requirement* (SCR). If its capital falls below the SCR level, an insurance company should, at minimum, deliver to the supervisor a plan to restore capital to above the SCR level. The supervisor might require the insurance company to take particular measures to correct the situation. The MCR is regarded as an absolute minimum level of capital. If capital drops below the MCR level, supervisors may assume control of the insurance company and prevent it from taking new business. It might force the insurance company into liquidation, transferring its policies to another company. The MCR is typically between 25% and 45% of the SCR.

There are two ways to calculate the SCR: the standardized approach and the internal models approach. The internal models approach involves a VaR calculation with a one-year time horizon and a 99.5% confidence limit. (The confidence level is therefore less than the 99.9% confidence level used in Pillar 1 of Basel II.) Longer time horizons with lower confidence levels are also allowed when the protection provided is considered equivalent. The SCR involves a capital charge for investment risk, underwriting risk, and operational risk. Investment risk is subdivided into market risk and credit risk. Underwriting risk is subdivided into risk arising from life insurance, non-life insurance (i.e., property and casualty), and health insurance.

Capital should be adequate to deal with large adverse events. Examples of the events considered in quantitative impact studies are:

1. A 32% decrease in global stock markets
2. A 20% decrease in real estate prices
3. A 20% change in foreign exchange rates
4. Specified catastrophic risk scenarios affecting property and casualty payouts
5. Health care costs increasing by a factor times the historical standard deviation of costs
6. A 10% increase in mortality rates
7. A 25% decrease in mortality rates
8. A 10% increase in expenses

The internal models are required to satisfy three tests. The first is a statistical quality test. This is a test of the soundness of the data and methodology used in calculating VaR. The second is a calibration test. This is a test of whether risks have been assessed in accordance with a common SCR target criterion. The third is a use test. This is a test of whether the model is genuinely relevant to and used by risk managers.

There are three types of capital in Solvency II. Tier 1 capital consists of equity capital, retained earnings, and other equivalent funding sources. Tier 2 capital consists of liabilities that are subordinated to policyholders and satisfy certain criteria concerning their availability in wind-down scenarios. Tier 3 capital consists of liabilities that are subordinated to policyholders and do not satisfy these criteria. Similarly to Basel II, the amount of capital that must be Tier 1, Tier 1 plus Tier 2, and Tier 1 plus Tier 2 plus Tier 3 is specified.

Summary

This chapter has provided an overview of capital requirements for banks up to 2007. The way in which regulators calculate the minimum capital a bank is required to hold has changed dramatically since the 1980s. Prior to 1988, regulators determined capital requirements by specifying minimum ratios for capital to assets or maximum ratios for assets to capital. In the late 1980s, both bank supervisors and the banks themselves agreed that changes were necessary. Derivatives trading was increasing fast. Also, banks were competing globally and it was considered important to create a level playing field by making regulations uniform throughout the world.

The 1988 Basel Accord assigned capital for credit risk for both on- and off-balance-sheet exposures. This involved calculating a risk-weighted asset for each item. The risk-weighted assets for an on-balance-sheet loan were calculated by multiplying the principal by a risk weight for the counterparty. In the case of derivatives such as swaps, banks were first required to calculate credit equivalent amounts. The risk-weighted assets were obtained by multiplying the credit equivalent amount by a risk weight for the counterparty. Banks were required to keep capital equal to 8% of the total risk-weighted assets. In 1995, the capital requirements for credit risk were modified to incorporate netting. As a result of an amendment introduced in 1996, the Accord was changed to include a capital charge for market risk. Sophisticated banks could base the capital charge on a value-at-risk calculation.

Basel II was proposed in 1999 and implemented by many banks in about 2007. It led to no immediate change to the capital requirement for market risk. Credit risk capital was calculated in a more sophisticated way than previously to reflect either (a) the credit ratings of obligors or (b) estimates made by the bank in conjunction with a default correlation parameter specified by regulators. In addition, there was a capital charge for operational risk.

Solvency II is a regulatory framework for insurance companies that was implemented by the European Union in 2016. It prescribes minimum capital levels for investment risk, underwriting risk, and operational risk. The structure of Solvency II is broadly similar to Basel II.

Further Reading

Bank for International Settlements. "Basel II: International Convergence of Capital Measurement and Capital Standards." June 2006, www.bis.org.

Gordy, M. B. "A Risk-Factor Model Foundation for Ratings-Based Bank Capital Ratios." *Journal of Financial Intermediation* 12 (2003): 199–232.

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Vasicek, O. "Probability of Loss on a Loan Portfolio." Working Paper, KMV, 1987. (Published in *Risk* in December 2002 under the title "Loan Portfolio Value.")

Practice Questions and Problems (Answers at End of Book)

25.1 "When a steel company goes bankrupt, other companies in the same industry benefit because they have one less competitor. But when a bank goes bankrupt other banks do not necessarily benefit." Explain this statement.

25.2 "The existence of deposit insurance makes it particularly important for there to be regulations on the amount of capital banks hold." Explain this statement.

25.3 An interest rate swap involves the exchange of a fixed rate of interest for a floating rate of interest with both being applied to the same principal. The principals are not exchanged. What is the nature of the credit risk for a bank when it enters into a five-year interest rate swap with a notional principal of $100 million? Assume the swap is worth zero initially.

25.4 In a currency swap, interest on a principal in one currency is exchanged for interest on a principal in another currency. The principals in the two currencies are exchanged at the end of the life of the swap. Why is the credit risk on a currency swap greater than that on an interest rate swap?

25.5 A four-year interest rate swap currently has a negative value to a financial institution. Is the financial institution exposed to credit risk on the transaction? Explain your answer. How would the capital requirement be calculated under Basel I?

25.6 Estimate the capital required under Basel I for a bank that has the following transactions with a corporation. Assume no netting.

(a) A nine-year interest rate swap with a notional principal of $250 million and a current market value of −$2 million.

(b) A four-year interest rate swap with a notional principal of $100 million and a current value of $3.5 million.

(c) A six-month derivative on a commodity with a principal of $50 million that is currently worth $1 million.

25.7 What is the capital required in Problem 25.6 under Basel I assuming that the 1995 netting amendment applies?

25.8 All the derivatives transactions a bank has with a corporate client have a positive value to the bank. What is the value to the bank of netting provisions in its master agreement with the client?

25.9 Explain why the final stage in the Basel II calculations for credit risk (IRB), market risk, and operational risk is to multiply by 12.5.

25.10 What is the difference between the trading book and the banking book for a bank? A bank currently has a loan of $10 million to a corporate client. At the end of the life of the loan, the client would like to sell debt securities to the bank instead of borrowing. How does this potentially affect the nature of the bank's regulatory capital calculations?

25.11 Under Basel I, banks do not like lending to highly creditworthy companies and prefer to help them issue debt securities. Why is this? Do you think this changed as a result of Basel II?

25.12 Banks sometimes use what is referred to as regulatory arbitrage to reduce their capital. What do you think this means?

25.13 Equation (25.9) gives the formula for the capital required under Basel II. It involves four terms being multiplied together. Explain each of these terms.

25.14 Explain the difference between the simple and the comprehensive approach for adjusting capital requirements for collateral.

25.15 Explain the difference between the standardized approach, the IRB approach, and the Advanced IRB approach for calculating credit risk capital under Basel II.

25.16 Explain the difference between the basic indicator approach, the standardized approach, and the advanced measurement approach for calculating operational risk capital under Basel II.

25.17 Suppose that the assets of a bank consist of $200 million of retail loans (not mortgages). The PD is 1% and the LGD is 70%. What are the risk-weighted assets under the Basel II IRB approach? What are the Tier 1 and Tier 2 capital requirements?

25.18 Section 11.10 discusses how statistics can be used to accept or reject a VaR model. Section 25.6 discusses guidelines for bank supervisors in setting the VaR multiplier m_c. It explains that, if the number of exceptions in 250 trials is five or more, then m_c is increased. What is the chance of five or more exceptions if the VaR model is working well?

Further Questions

25.19 Why is there an add-on amount in Basel I for derivatives transactions? "Basel I could be improved if the add-on amount for a derivatives transaction depended on the value of the transaction." How would you argue this viewpoint?

25.20 Estimate the capital required under Basel I for a bank that has the following transactions with another bank. Assume no netting.

(a) A two-year forward contract on a foreign currency, currently worth \$2 million, to buy foreign currency worth \$50 million

(b) A long position in a six-month option on the S&P 500. The principal is \$20 million and the current value is \$4 million.

(c) A two-year swap involving oil. The principal is \$30 million and the current value of the swap is –\$5 million.

What difference does it make if the netting amendment applies?

25.21 A bank has the following transaction with a AA-rated corporation

(a) A two-year interest rate swap with a principal of \$100 million that is worth \$3 million

(b) A nine-month foreign exchange forward contract with a principal of \$150 million that is worth –\$5 million

(c) A long position in a six-month option on gold with a principal of \$50 million that is worth \$7 million

What is the capital requirement under Basel I if there is no netting? What difference does it make if the netting amendment applies? What is the capital required under Basel II when the standardized approach is used?

25.22 Suppose that the assets of a bank consist of \$500 million of loans to BBB-rated corporations. The PD for the corporations is estimated as 0.3%. The average maturity is three years and the LGD is 60%. What is the total risk-weighted assets for credit risk under the Basel II IRB approach? How much Tier 1 and Tier 2 capital is required? How does this compare with the capital required under the Basel II standardized approach and under Basel I?